Silicon-Based Optical Phased Array Using Electro-Optic *p-i-n* Phase Shifters

Geumbong Kang, Seong-Hwan Kim, Jong-Bum You, Dae-Seong Lee, Hyeonho Yoon, Yun-Gi Ha, Jong-Hun Kim, Dong-Eun Yoo, Dong-Wook Lee, Chan-Hyun Youn, Kyoungsik Yu, and Hyo-Hoon Park

***Abstract*— We present a 1 × 16 silicon optical phased array using electro-optic phase shifters to attain high-speed operation with low power consumption. The phase shifters are constructed using a *p-i-n* junction structure with the *p*- and *n*-doped regions formed on both sides of the *i*-region. The *i*-region width of the phase shifter is optimized considering the power consumption for phase-tuning and the propagation loss in the phase shifter. The fabricated *p-i-n* phase shifter exhibits a fast operating speed of 20 MHz and a low phase-tuning power of 1.7 mW/ π. From a 1-D optical phased array integrated with 2-μm-pitch grating radiators, a wide beam-steering range of 45° is attained along the transversal direction at a 1.55 μm wavelength. Average power consumption for the beam-forming operation of the 1 × 16 OPA is measured to be 39.6 mW and the average transition time to steer the beam is 24 ns.**

***Index Terms*— Optical phased array, silicon photonics, electro-optic, *p-i-n* phase shifter.**

I. Introduction

THE optical phased array (OPA) has emerged as a promising component for light detection and ranging (LiDAR) systems, based on the inherent potential of its semiconductor components [1]–[3]. Compared to the mechanical laser scanner employed in commercialized LiDARs, the OPA offers compact integration, low power consumption and stable operation. Also, the silicon-based OPA can open the way to the low-cost mass production of LiDAR components.

In the Si-based OPA, the phase shifter is a key element for beam-steering, which is operated by controlling the refractive index (RI) of the waveguides. Most of the OPAs reported to date have adopted a thermo-optic (TO) phase shifter that controls the RI using the silicon thermo-optic effect. Various heaters such as a doped silicon heater [1], [3], direct *p-i-n* heater [4], and indirect metal heater [5] have been explored for the TO phase shifter. These TO phase shifters provided low optical loss without bias voltage dependency. However, the TO phase shifters possess inherent drawbacks in operation speed and power consumption due to the use of heat. To change the RI, it requires high-temperature heating of the silicon or insulator waveguide up to several hundred °C [1]. Using such heating structures, the operating speed is limited to the megahertz level [2] and power consumption reaches several tens of milliwatts per channel for full beam-steering [1], [4]. The power consumption issue becomes more serious when the scale of OPA is increased to reduce beam divergence.

In contrast to the TO effect, the electro-optic (EO) effect can provide fast operation and low power consumption in silicon waveguide devices [6], since the RI is controlled by the carrier concentration, based on the free carrier plasma dispersion effect [7]. Several works [8]–[10] have implemented the EO phase shifters in silicon OPAs, but the design issues of the phase shifters were not reported in detail. These works focused on the architecture of a 2-D OPA [8], [9] or system demonstration [10].

In our previous work [11], an EO phase shifter was employed with a TO tunable radiator for 2-D beam-steering from a 1-D OPA, but the reported design and performance of the OPA mainly focused on the tunable radiator. For wide beam-steering, the 1-D OPA is absolutely advantageous, compared to the 2-D OPA, since allowed space in the radiator array to attain a wide diffraction angle is too narrow to integrate the elementary components of the waveguide line, the phase controller and the radiator in the case of 2-D configuration. Using the 1-D OPA, the longitudinal beam-steering can be achieved by employing other schemes, such as wavelength tuning from the source [2], [3] and index tuning in the radiator [11].

In this letter, we propose an optimized design for the phase shifter of a 1-D OPA. A detailed analysis of the power consumption for phase-tuning and propagation loss are conducted on the phase shifter structure. The power consumption and speed for beam forming and steering are evaluated through the operation of the OPA.

II. Design and Simulation

Figure 1(a) shows the layout of the designed 1-D OPA. It consists of an input grating coupler, multi-mode

Manuscript received June 24, 2019; revised August 29, 2019; accepted August 30, 2019. Date of publication September 30, 2019; date of current version November 12, 2019. This work was supported in part by the Civil-Military Technology Cooperation Program under Grant N04140260, in part by the Future Semiconductor Device Technology Development Program through the Ministry of Trade, Industry and Energy (MOTIE) and the Korea Semiconductor Research Corporation (KSRC) under Grant 10085592, and in part by the Brain Korea 21 Plus Project of School of Electrical Eng. KAIST in 2019. *(Corresponding author: Hyo-Hoon Park.)*

G. Kang, S.-H. Kim, D.-S. Lee, H. Yoon, Y.-G. Ha, J.-H. Kim, C.-H. Youn, K. Yu, and H.-H. Park are with the School of EE, KAIST, Daejeon 34141, South Korea (e-mail: rkdrmaqhd@kaist.ac.kr; ksh113@kaist.ac.kr; daeseonglee@kaist.ac.kr; exhyho@kaist.ac.kr; yungi.ha@kaist.ac.kr; jhkim09@kaist.ac.kr; chyoun@kaist.ac.kr; ksyu@kaist.ac.kr; parkhh@kaist.ac.kr).

J.-B. You, D.-E. Yoo, and D.-W. Lee are with the National NanoFab Center (NNFC), Daejeon 34141, South Korea (e-mail: jbyou@nnfc.re.kr; deyoo@nnfc.re.kr; dwlee@nnfc.re.kr).

Color versions of one or more of the figures in this letter are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/LPT.2019.2939550

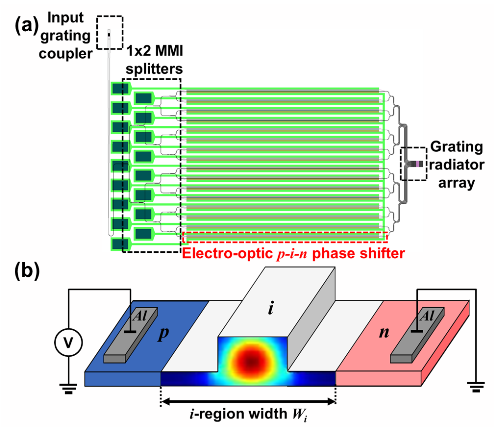


Fig. 1. (a) Layout of designed optical phased array using electro-optic phase shifters and grating radiators. (b) Schematic of the designed structure of the electro-optic p-i-n phase shifter.

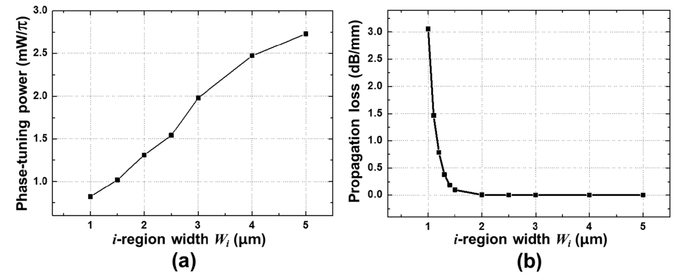


Fig. 2. Simulated (a) phase-tuning power and (b) propagation loss as a function of W_i in the p-i-n phase shifter.

interferometer (MMI) power splitters, EO phase shifters and grating radiators with an array of 16 channels. The input grating coupler is designed for coupling from a fiber-tailed 1550 nm light source. To split the input light into 16 channels, four stages of 1×2 MMI are configured. Using the multi stages of the MMI splitters, the channels are paned-out to the phase shifters with a pitch of 50 μm. This pitch is determined to sufficiently lessen the bending loss, despite that the minimum pitch between each phase shifter without the pan-out region is close to 10 μm with our linear configuration of the electrodes contacted through the oxide.

The EO phase shifters are designed with p-i-n junctions, in which the p- and n-doped regions are placed on both sides of the intrinsic region containing each waveguide core. Following the phase shifters, the channels are paned-in to reach compactly arrayed radiators. The pan-in waveguides are configured with multi-stages similar to the pan-out stages, while maintaining separated channels, as shown in Fig. 1(a). Such multi-staged pan-out and pan-in channels were devised to equalize channel length, as a trial to remove the phase difference due to the path difference. The grating radiator array was designed with a uniform pitch of 2 μm which can provide beam-steering of over 45° with a zeroth-order single beam.

The detailed structure of the EO phase shifter is shown in Fig. 1(b). The p-i-n junction and electrodes are placed along a straight waveguide, like a typical 1-D OPAs [2], [3]. The waveguide was designed with a width of 500 nm, a height of 220 nm and a slab height of 70 nm. The p- and n-doped regions were formed on the slab with a distance W_i which is defined as the i-region width. The i-region width W_i was determined by considering the phase-tuning efficiency and propagation loss of the p-i-n waveguide, which is described below. For electrical contact, aluminum pads were placed on top of the doped regions. The length of the p-i-n phase shifter was set to 1 mm to attain the phase shift required for beam-steering of the OPA with a low voltage level, to retain a compact device size.

To ensure low power consumption and low propagation loss in the phase shifter, we analyzed the effect of W_i using numerical simulations. When W_i decreases, closing the doped regions to the waveguide core, the resistance decreases. Thus, the applied voltage needed to supply the same carrier concentration in the waveguide core decreases, and the power consumption needed for phase tuning decreases [12]. However, if W_i becomes too narrow, optical loss increases severely since the doped region makes inroads into the evanescent field of the optical mode. Therefore, W_i should be optimized considering this trade-off relationship.

Figure 2 shows the simulated results of the power consumption and optical loss of the p-i-n phase shifter with respect to W_i. For this simulation, we used Lumerical DEVICE and MODE solution. The carrier distribution according to W_i and the bias voltage were calculated first using the DEVICE simulation. Then the carrier distribution was applied to the MODE simulation to calculate the effective RI of the fundamental mode. W_i varied from 1 μm to 5 μm and surrounding p- and n-doped regions were set up with a uniform doping concentration of 5×10^{20} cm^{-3}. In Fig. 2(a), the phase-tuning power means the power consumption for π-phase shift, represented with a unit of mW/π. As shown in Fig. 2(a), when W_i increases, the phase-tuning power increases gradually. Note that, in the large W_i region consuming high electrical high power, the heating effect could be added. Since the TO effect increases the RI and diminishes the EO effect [2], more power may be required than the value in Fig. 2(a) in the large W_i region. Thus, considering only the tuning power, it is advantageous to narrow W_i down. However, when W_i decreases the propagation loss of the optical field can increase, because the evanescent field penetrates more into the p- and n-doped regions. In the simulated result in Fig. 2(b), the loss shows an abrupt increase below 1.5 μm. Above 1.5 μm, the loss reaches almost zero level, which corresponds to the loss of the intrinsic waveguide.

In Fig. 2(b), the propagation loss under zero bias is presented. When bias is applied, the loss increases almost proportionally to the current [6], i.e., the amount of the phase shift. The additional loss for the 2π-phase shift was calculated to be in the range of $2.9 \sim 3.2$ dB, almost regardless of W_i. Considering the trade-off relationship between the phase-tuning power and the propagation loss with the additional loss for phase shift, the optimal range of W_i would be within $1.3 \sim 2.5$ μm. Within this range, we selected a W_i with 2.5 μm to demonstrate the OPA, considering fabrication errors. In the fabrication process, misalignment of the mask and the lateral distribution of dopants may have the effect of reducing the width of the intrinsic region more than the condition given in the simulation. To avoid the possibility of reaching the critical width showing a serious loss, we left enough margin in the selection of W_i.

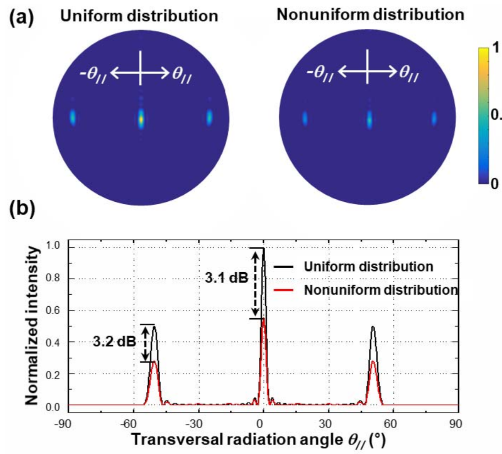


Fig. 3. Simulated (a) far-field patterns and (b) intensity distribution along the transversal radiation direction, with uniform and random distributions of optical intensities in 16 channels.

The optical field passed through the phase shifter can fluctuate during phase control, since the optical loss varies with the injected current. Thus the radiated beam could not form a clear zeroth-order diffraction pattern although the phases are well controlled. To inspect this problem, we compared the radiation characteristics of the 1×16 OPA through FDTD simulation for two cases; supplying a uniform or a nonuniform intensity of the optical field to the radiators. For the nonuniform field, a random distribution between 0.5 and 1 was given in the relative optical intensity supplied to each radiator. This range of relative intensity distribution was determined by considering that the additional propagation loss for the 2π-phase shift was simulated to be about 3 dB, as explained in Fig. 2.

Figure 3(a) shows the simulated far-field patterns of the in-phase state for uniform and nonuniform distribution. Here, the in-phase state is the state where the phase difference between adjacent channels becomes zero ($\Delta\phi = 0°$), at which a zeroth-order main beam is formed at the center ($\theta_{//} = 0°$). For both distributions, clear far-field patterns are formed, while the zeroth- and first-order diffraction beams show weaker intensities for the nonuniform distribution. The transversal and longitudinal beam divergences and the side-lobe suppression ratio remain as 3.3°, 6.2° and 12.6 dB, respectively.

The difference in intensity between the two distributions is 3.1 dB for the zeroth-order and 3.2 dB for the first-order beams, as shown in Fig. 3(b). Since the intensity difference of these beams is almost the same as the additional optical loss of the phase shifter during the 2π-phase shift, it can be seen that the nonuniform distribution of the optical intensity does not obstruct the beam-forming of the OPA. Note that we also observed a similar tendency in the π-phase state ($\Delta\phi = \pi$), where two symmetrical beams are formed, reaching the maximum angle attainable with a single beam.

These results indicate that single beam-steering can be possible even though the intensities of the optical field are varied during the phase control of the EO phase shifter. A similar consensus was also reported in other OPA works using EO phase shifters [8], [9], where the effects of the nonuniform E-field were analyzed with different conditions.

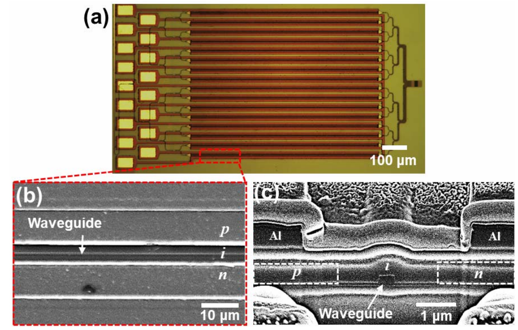


Fig. 4. (a) Microscope image of whole OPA, (b) SEM and (c) cross-sectional FIB images of EO phase shifters of fabricated devices.

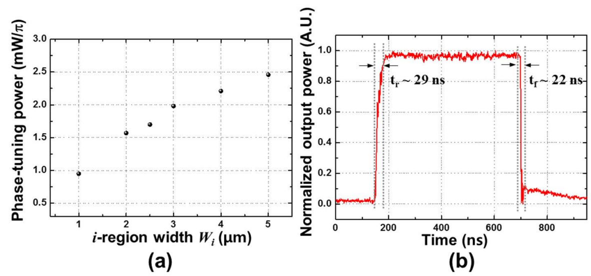


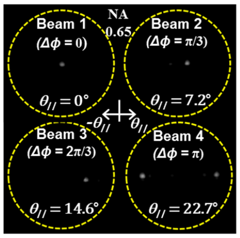
Fig. 5. Experimental data of (a) phase-tuning power with variation of the *i*-region width *Wi* in the *p-i-n* phase shifter, measured with MZI test devices. (b) Temporal response of a MZI test device showing the modulation speed of the electro-optic *p-i-n* phase shifter.

III. Fabrication and Characterization

The OPA was fabricated on an 8-inch SOI wafer with a top silicon layer of 220 nm and a buried oxide layer of 2 μm, using CMOS compatible processes. A KrF scanner with a 248 nm wavelength laser was used for the photolithography. The SIMS analysis indicated that the *p*- and *n*-doped regions had doping concentrations of 3×10^{20} cm^{-3} and 2.8×10^{20} cm^{-3}, respectively. The footprint of the fabricated OPA was 1.1 mm × 1.6 mm. Figures 4(a), (b) and (c) show, respectively, the microscope image of the fabricated OPA, SEM image of the phase shifter, and the cross-sectional FIB image of the phase shifter.

Figure 5 shows the experimental results obtained with the fabricated phase shifter. The electrical characteristics of the phase shifter were analyzed with Mach-Zehnder interferometer (MZI) test devices. Figure 5(a) shows the measured phase-tuning power of the MZIs with different *i*-region widths W_i. TE-polarized light with a wavelength of 1550 nm was used as the input source. The tuning power increases with increasing W_i, as shown in the simulation result of Fig. 2(a). At 2.5 μm width, selected to demonstrate the OPA, the tuning power was measured to be 1.7 mW/π and the 2π-phase shift was achieved at 1.1 V. In addition, the rising and falling times were measured using a MZI test device to be 29 and 22 ns, with a peak-to-peak voltage of 1.1 V, as shown in Fig. 5(b). The measured temporal response corresponds to a modulation speed of 20 Mbps.

Considering the *p-n* phase shifter, the tuning power and the speed can be much improved due to the reverse bias operation, as demonstrated in recent work [10]. The work showed a tuning power of 2 μW/2π which is orders of magnitude less than our work (1.7 mW/π). However, in general, the *p-n* structure



(a)

	Beam 1	Beam 2	Beam 3	Beam 4
V_1	1.02 V	1.08 V	0.75 V	0.91 V
V_6	1.05 V	0.89 V	0.96 V	1.08 V
V_{11}	0.97 V	1.01 V	1.06 V	0.84 V
V_{16}	0.70 V	1.03 V	0.74 V	0.88 V
Total power	40.3 mW	37.2 mW	38.7 mW	42.1 mW

Transition time 24 ns 26 ns 22 ns

(b)

Fig. 6. (a) Measured far-field patterns of the OPA showing 1-D beam-steering to transversal directions and (b) the bias voltages applied to 4 samples of the phase shifters to steer the beam.

shows a larger voltage-length product ($V_\pi L$) over an order of magnitude and it requires a larger bias voltage or a longer length [7], [13]. Thus, the *p-n* structure may require a larger chip area, even the pitch of the *p-i-n* structure also extends due to the insertion of *i*-region and for current isolation. In addition, the *p-n* waveguide suffers a fundamental loss from dopants. Nevertheless, the previous work [10] accomplished relatively low loss (2.4 dB in average) close to this work, even the detailed structure and bias condition were not released. The disadvantages of the *p-n* structure can be considerably diminished, when III-V materials are employed [13].

Figure 6(a) shows the far-field patterns measured from the OPA. The emitting beam from the OPA was collected by a Fourier image system with a high NA (0.65) objective lens. To minimize the side field distribution, the voltage biases of the phase shifters were found using the hill-climbing algorithm [2]. With our program, it required about 80 times of iteration for whole channels to obtain a clear beam.

Figure 6(a) indicates that the maximum steering range attainable with a single beam is $2\theta_{//} = 45.4°$ from the appearance of two beams caused by the shift of one of the first-order diffraction beams, as seen in beam 4. This state corresponds to the π-phase state, which is ideally attainable with a phase difference of $\Delta\phi = \pi$ for each radiator. The transversal and longitudinal beam divergence and side-lobe suppression ratio were measured to be 3.4°, 5.9° and 8.2 dB, respectively, showing a slight degradation in the side-lobe suppression, compared to the simulation. The loss data of element devices are presented in previous work [11].

For each beam in Fig. 6(a), the steered angle $\theta_{//}$ in the transversal direction and the ideal phase difference $\Delta\phi$ corresponding to $\theta_{//}$ are noted. To check the phase relation during beam-steering, the bias voltages applied to each of the phase shifters were analyzed. Figure 6(b) shows the variation in bias voltages during beam-steering for 4 samples among 16 phase shifters. The data shows that the bias voltages are almost randomly set for each phase shifter during beam-steering. This result indicates that, even though we designed a symmetric pan-out and pan-in structure, providing equal channel lengths, the phases supplied to the radiators does not consistently change, unlike what is expected from the ideal condition. The tendency can be attributed to deviations in the geometry and the doping concentration of the fabricated waveguides.

In Fig. 6(b), we also provide the total power consumption of the 16 phase shifters used to form each beam, defined by $P_{total} = \sum_{n=1}^{16} V_n I_n$. The data of V_n and I_n for the n-th phase shifter were obtained from the OPA chip used for the measurement of Fig. 6(a). The total power consumption for each beam-forming shows a small amount of variation, within 10% of the average value of 39.6 mW. In addition, the transition times to shift the beam direction are provided below the table in Fig. 6(b). The transit time was determined by the response time of the MZI test device which measured from the voltage corresponding to the largest bias voltage among the 16 phase shifters for each beam-forming. The average transition time obtained from this method is 24 ns. Since the power consumption and transition time are measured based on the actual beam-scanning operation of the OPA, the results could be practical data for LiDAR system operation.

IV. Conclusion

In conclusion, we demonstrated a silicon-based 1×16 optical phased array using electro-optic *p-i-n* phase shifters. The effects of the intrinsic (*i*) region width of the *p-i-n* phase shifters was analyzed to achieve low phase-tuning power and low propagation loss. Experimentally, we achieved a low phase-tuning power of 1.7 mW/π with a modulation speed of 20 MHz from a phase shifter device and a wide beam-steering range of 45.4° from an OPA. The average power required to operate 16 channels during beam-steering was measured to be 39.6 mW and the average transition time to steer each beam was 24 ns. The electro-optic OPA offers low power consumption, high-speed beam-steering and wide beam-steering range, which are attractive for application in LiDAR systems.

References

[1] J. Sun, E. Timurdogan, A. Yaacobi, E. S. Hosseini, and M. R. Watts, "Large-scale nanophotonic phased array," *Nature*, vol. 493, no. 7431, pp. 195–199, Jan. 2013.

[2] J. C. Hulme *et al.*, "Fully integrated hybrid silicon two dimensional beam scanner," *Opt. Express*, vol. 23, no. 5, pp. 5861–5874, Mar. 2015.

[3] D. N. Hutchison *et al.*, "High-resolution aliasing-free optical beam steering," *Optica*, vol. 3, no. 8, pp. 887–890, Aug. 2016.

[4] J. C. Hulme *et al.*, "Fully integrated hybrid silicon free-space beam steering source with 32-channel phased array," in *Proc. SPIE Photon. West*, San Francisco, CA, USA, vol. 8989, Mar. 2014.

[5] D. Kwong *et al.*, "On-chip silicon optical phased array for two-dimensional beam steering," *Opt. Lett.*, vol. 39, no. 4, pp. 941–944, Feb. 2014.

[6] K. Yamada, T. Tsuchizawa, T. Watanabe, H. Fukuda, H. Shinojima, and S. Itabashi, "Applications of low-loss silicon photonic wire waveguides with carrier injection structures," in *Proc. IEEE-GFP*, Tokyo, Japan, Sep. 2007, pp. 116–118.

[7] G. T. Reed, G. Mashanovich, F. Y. Gardes, and D. J. Thomson, "Silicon optical modulators," *Nature Photon.*, vol. 4, pp. 518–526, Jul. 2010.

[8] F. Aflatouni, B. Abiri, A. Rekhi, and A. Hajimiri, "Nanophotonic projection system," *Opt. Express*, vol. 23, no. 16, pp. 21012–21022, Aug. 2015.

[9] C. Zhao, C. Peng, and W. Hu, "Blueprint for large-scale silicon optical phased array using electro-optical micro-ring pixels," *Sci. Rep.*, vol. 7, Dec. 2017, Art. no. 17727.

[10] C. V. Poulton *et al.*, "Long-range LiDAR and free-space data communication with high-performance optical phased arrays," *IEEE J. Sel. Topics Quantum Electron.*, vol. 25, no. 5, Sep./Oct. 2019, Art. no. 7700108.

[11] S.-H. Kim *et al.*, "Thermo-optic control of the longitudinal radiation angle in a silicon-based optical phased array," *Opt. Lett.*, vol. 44, no. 2, pp. 411–414, Jan. 2019.

[12] C. E. Png, S. P. Chan, S. T. Lim, and G. T. Reed, "Optical phase modulators for MHz and GHz modulation in silicon-on-insulator (SOI)," *J. Lightw. Technol.*, vol. 22, no. 6, pp. 1573–1582, Jun. 2004.

[13] W. Xie *et al.*, "Heterogeneous silicon photonics sensing for autonomous cars [Invited]," *Opt. Express*, vol. 27, no. 3, pp. 3642–3663, Feb. 2019.